DreamVLA

From Generated Human Motions to Humanoid Vision-Language-Action Policies

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ember-center-berkeley.github.io/DreamVLA

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Generated motion → residual expert → corrective recovery → vision-language-action policy

01 Generate demonstrations

Replace repeated task-specific teleoperation with generated motion and physically executed expert demonstrations.

1 Generate and retarget

Task text + contact constraints → OmniControl human motion → OmniRetarget G1 references. Augment with HumanoidMimicGen.

2 Adapt contact with residual RL

SONIC supplies a whole-body motion prior. A privileged residual expert co-tracks the robot and task object under randomized physics.

3 Keep successful expert rollouts

Record egocentric RGB, proprioception, executed tokens, hand commands, and language in LeRobot format. Initialize GR00T on accepted episodes.

Executed token = SONIC token + learned residual

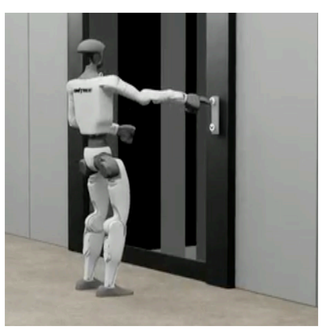
Four simulated G1 interactions



Pick from platform



Open drawer



Press button



Walk and pick

02 Recover from learner errors

Offline demonstrations cover expert states. Learner errors alter contacts and move the VLA away from that distribution.

1 Roll out the current VLA

Execute images + proprioception + language. Save simulator states along the learner trajectory.

↓

2 Restore a failed learner state

Sample a time uniformly from the failed rollout. Restore that state and resume the same reference suffix.

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3 Collect an expert continuation

The privileged expert attempts recovery. Keep the corrective trajectory only when it completes the task.

↓

4 Aggregate and fine-tune

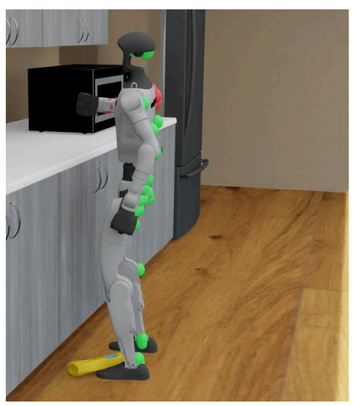
Add accepted continuations to the dataset; fine-tune GR00T and repeat for three DAgger rounds.

Student deployment

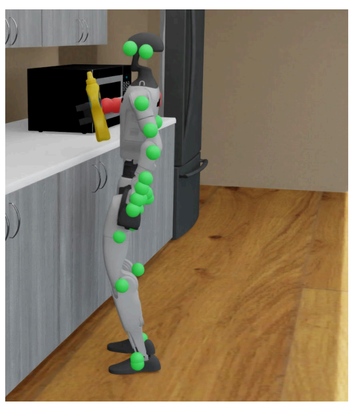
RGB + robot state + instruction → token / hand actions → frozen SONIC decoder. No privileged object state or reference input.

Residual adaptation at contact

Nominal SONIC

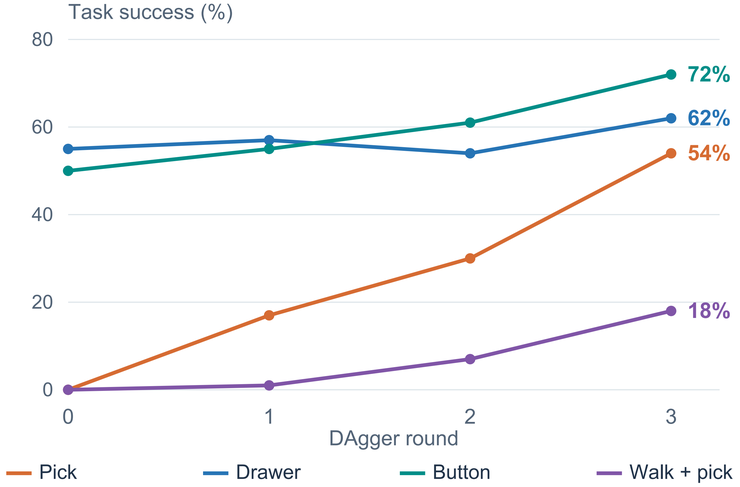


Residual SONIC



Matched final frames from the paper: nominal tracking loses the bottle; residual execution retains it.

03 Measure closed-loop success



100 evaluation rollouts per task; paper Table 2.

Corrective data vs. more offline data

Task	Offline	DAgger-3	Offline control
Pick	0%	**54%**	0%
Drawer	55%	**62%**	54%
Button	50%	**72%**	52%
Walk + pick	0%	**18%**	0%

Walking retained episodes, base → R3: 200 → 314 → 416 → 535. Successful corrections are filtered after rollout attempts.

Generated-data scale: offline only

Episodes	Pick	Drawer	Button	Walk + pick
10	0%	4%	0%	0%
50	0%	15%	15%	0%
200	0%	55%	50%	0%

Main finding

More offline data improves drawer and button success. DAgger raises pick success from 0% to 54% in this study.

Scope: simulation only; pretrained motion, control, and VLA backbones remain in use.

Built on DreamControl / OmniControl · OmniRetarget · HumanoidMimicGen · SONIC · GR00T. DAgger: Ross et al., AISTATS 2011.

Results: supplied paper, Tables 2–4